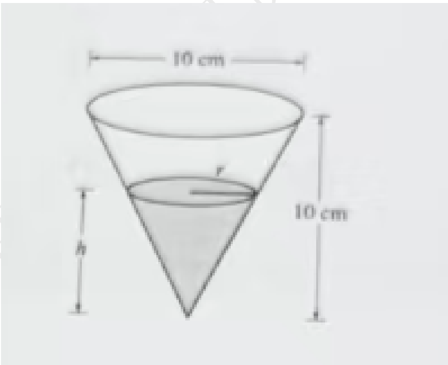
贺昱超练习卷

2026 年 7 月 5 日

Related Rates Practice

3. [20 points]



A container has the shape of an open right circular cone, as shown in the figure above. The height of the container is 10 cm and the diameter of the opening is 10 cm. Water in the container is evaporating so that its depth h is changing at the constant rate of

$$\frac{dh}{dt} = -\frac{3}{10} \text{ cm/hr}.$$

Note: The volume of a cone of height h and radius r is given by

$$V = \frac{1}{3}\pi r^2 h.$$

(a) Find the volume V of water in the container when $h = 5$ cm. Indicate units of measure.

(b) Find the rate of change of the volume of water in the container, with respect to time, when $h = 5$ cm. Indicate units of measure.

Similar Practice Problems

1. [20 points]

A water tank has the shape of an open right circular cone. The height of the tank is 12 cm and the diameter of the opening is 8 cm. Water is being poured into the tank so that the water depth h is increasing at the constant rate

$$\frac{dh}{dt} = \frac{1}{4} \text{ cm/min.}$$

Note: The volume of a cone of height h and radius r is

$$V = \frac{1}{3}\pi r^2 h.$$

(a) Find the volume of water in the tank when $h = 6$ cm.

(b) Find the rate of change of the volume of water in the tank when $h = 6$ cm. Indicate units of measure.

2. [20 points]

An inverted cone-shaped container has height 15 cm and radius 6 cm at the top. Water is being poured into the container so that the depth of the water is increasing at the rate

$$\frac{dh}{dt} = 0.2 \text{ cm/s.}$$

(a) Express the radius r of the water surface in terms of the water depth h.

(b) Find the volume of water when $h = 10$ cm.

(c) Find the rate of change of the volume when $h = 10$ cm. Indicate units of measure.

3. [20 points]

A conical cup has height 9 cm and diameter 6 cm at the top. Water is being added to the cup. The depth of the water increases at the constant rate

$$\frac{dh}{dt} = 0.5 \text{ cm/hr.}$$

(a) Find a formula for the volume V of water in terms of h only.

(b) Find the volume of water when $h = 3$ cm.

(c) Find the rate of change of the volume when $h = 3$ cm. Indicate units of measure.

4. [20 points]

A funnel has the shape of an open right circular cone. The height of the funnel is 20 cm and the radius of the opening is 8 cm. Water is being poured into the funnel, and the depth of the water is increasing at the rate

$$\frac{dh}{dt} = 0.3 \text{ cm/s}.$$

(a) Write an equation relating the radius r of the water surface and the depth h.

(b) Find the volume of water in the funnel when $h = 10$ cm.

(c) Find the rate at which the volume of water is changing when $h = 10$ cm. Indicate units of measure.

5. [20 points]

A cone-shaped tank has height 18 cm and diameter 12 cm at the top. Water is being added to the tank so that the water depth increases at the constant rate

$$\frac{dh}{dt} = 0.4 \text{ cm/min}.$$

(a) Express the volume V of water as a function of h.

(b) Find the volume of water when $h = 9$ cm.

(c) Find $\frac{dV}{dt}$ when $h = 9$ cm. Indicate units of measure.

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